

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model for maximum friction = μmg	AO3.3	B1	$F_{\max} = \mu mg$ $= 0.85 \times 20 \times 9.8$ $= 166.6 \text{ N}$
	Makes an appropriate comparison	AO1.1a	M1	
	Explains clearly why crate remains stationary	AO2.4	E1	$150 < 166.6$ $\therefore$ crate does not move
(b)	Forms an equation by resolving vertically Condone one of sign error or cos error	AO3.1b	M1	$20g = R + 150\sin 15^\circ$ $R = 157.177 \text{ N}$
	Obtains correct reaction force	AO1.1b	A1	$F_{\max} = \mu \times 157.177$ $= 133.6 \text{ N}$
	Uses maximum friction = μR With 'their' reaction force Must identify maximum or limiting friction	AO1.1b	B1F	$150\cos 15^\circ = 145 \text{ N}$ $145 > 133.6$ $\therefore$ crate begins to move
	Compares $150\cos 15^\circ$ with 'their' maximum friction	AO1.1a	M1	
	Explains, using their values, why the crate begins to move.	AO2.4	E1F	
				Total 8 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Applies Newton's 2 nd Law to form a 3 term equation Award mark even if signs not correct	AO1.1a	M1	$F - 80 \times 10 = -80 \times 1.5$ $F - 800 = -120$ $F = 680 = 700(\text{N})$ to 1 sf
Obtains a correct 3 term equation.	AO1.1b	A1	
Obtains correct reaction	AO1.1b	A1F	

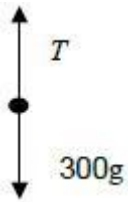
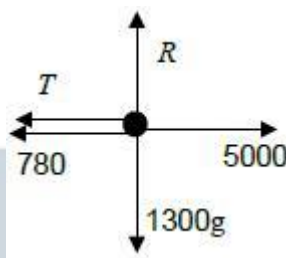
force. Must be given to 1 sf FT from incorrect 3 term equation provided M1 mark was awarded (condone omission of units)			
Total 3 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{dv}{dt} = 12t - 36t^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4 (12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r , with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3) dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^3 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds

marks have been awarded		
Total 8 marks		

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws correct force diagram for crate from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(i)	Draws correct force diagram for van from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(b)	Applies Newton's 2nd Law ($F = ma$) to the crate	AO3.4	M1	For crate $T - 300g = 300a$
	Applies Newton's 2nd Law ($F = ma$) to the van ($F = ma$ 'round the corner' scores 0)	AO3.4	M1	For van $5000 - T - 780 = 1300a$ $4220 - T = 1300a$
	Solve their simultaneous equations	AO1.1a	M1	$4220 - 300g = 1600a$
	Finds the value of a correctly AG	AO1.1b	A1	$a = 1280 \div 1600 = 0.80 \text{ m s}^{-2}$ (AG)
(c)	Uses $a = 0.80$ in either of their two equations in (b)	AO3.4	M1	$T = 300 \times 0.80 + 300g$
	Finds the correct value for T (condone omission of units) Possibly done in (b)	AO1.1b	A1	$= 3180$ $= 3200 \text{ (N) (2 sf)}$
(d)	Explains that the model could be refined by including air resistance	AO3.5c	E1	Resistance will increase with speed
				Total 9 marks

Q5.

Marking Instructions	AO	Marks	Typical Solution
Uses correct forces to form a moment equation (PI)	AO1.1a	M1	Take moments about C: $Mg \times 0.8 = 0.7 \times 24$
Obtains correct value	AO1.1b	A1	$M = 21$
Total 2 marks			

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Sums the forces given correctly	AO1.1b	B1	$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ $= 33\mathbf{i} - 11\mathbf{j}$
	Uses Pythagoras to find the magnitude of the vector and obtains correct magnitude (given to 3 sig figs)	AO1.1b	B1	$ \mathbf{F} = \sqrt{33^2 + (-11)^2}$ $= 34.8 \text{ N (3 sf)}$
	(ii) Uses trig expression with appropriate values	AO1.1a	M1	$\tan \theta = \frac{11}{33}$
	Obtains correct angle (given to nearest 0.1°)	AO1.1b	A1	$\theta = \tan^{-1}\left(\frac{1}{3}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\sin \theta = \frac{11}{\sqrt{1210}}$ $\theta = \sin^{-1}\left(\frac{11}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$ OR $\cos \theta = \frac{33}{\sqrt{1210}}$ $\theta = \cos^{-1}\left(\frac{33}{\sqrt{1210}}\right)$ $= 18.4^\circ \text{ (3 sf)}$
(b)	States negative of 'their' part (a)(i)	AO2.2a	B1F	$\mathbf{F}_4 = -33\mathbf{i} + 11\mathbf{j}$

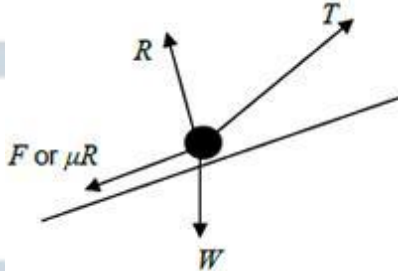
Total 5 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates two (or four) appropriate distances	AO3.1b	M1	$s_1 = \frac{1}{2} (6 + 10) \times 8 = 64 \text{ m}$
	Obtains correct distances	AO1.1b	A1	$s_2 = \frac{1}{2} \times 10 \times 2 = 10 \text{ m}$
	Obtains correct sum of 'their' distances	AO1.1b	A1F	$s_1 + s_2 = 64 + 10 = 74 \text{ m}$ OR $S_1 = 6 \times 8 = 48 \text{ m}$ $S_2 = \frac{1}{2} \times 4 \times 8 = 16 \text{ m}$ $S_3 = \frac{1}{2} \times 4 \times 2 = 4 \text{ m}$ $S_4 = \frac{1}{2} \times 6 \times 2 = 6 \text{ m}$ $S_1 + S_2 + S_3 + S_4 = 74 \text{ m}$
(b)	Finds difference of 'their' distances from part (a)	AO2.2a	B1F	$64 - 10 = 54 \text{ m}$
(c)	Calculates magnitude of acceleration	AO3.1b	M1	$a_{\max} = \frac{8}{4} = 2$
	Obtains correct resultant force	AO1.1b	A1	$F_{\max} = 800 \times 2 = 1600 \text{ N}$
(d)	Explains that abrupt changes and straight lines in the graph are unlikely in reality	AO3.5b	E1	Change of velocity is unlikely to result in abrupt changes I would expect to see curves on the graph
				Total 7 marks

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40^\circ$ Resolving horizontally

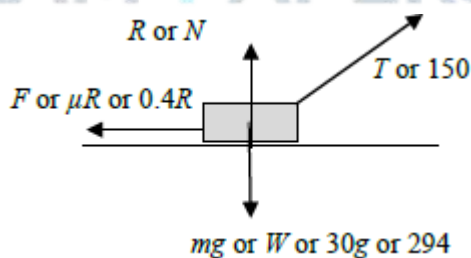
(allow consistent mixing of sin and cos)			$F = 50 \cos 40^\circ$
Obtains correct expressions for R and F	AO1.1b	A1	
States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50 \cos 40^\circ = \mu(8 \times 9.8 - 50 \sin 40^\circ)$ $\mu = \frac{50 \cos 40^\circ}{8 \times 9.8 - 50 \sin 40^\circ}$
Completes a rigorous argument that results with correct μ AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) AG
(b) (i) Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	

(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R , $8g \cos 5^\circ$ and $T \sin 40^\circ$ (or $T \cos 50^\circ$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8 \cos 5^\circ - T \sin 40^\circ$ $= 78.1 - T \sin 40^\circ$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T \cos 40^\circ - 8 \times 9.8 \sin 5^\circ - F = 8 \times 3$ $T \cos 40^\circ - 8 \times 9.8 \sin 5^\circ - 0.827 \dots \times (8 \times 9.8 \cos 5^\circ - T \sin 40^\circ) = 24$
	Forms a four term equation of motion parallel to the plane with correct terms	AO3.1b	M1	$T = \frac{24 + 8 \times 9.8 \sin 5^\circ + 0.827 \dots \times 8 \times 9.8 \cos 5^\circ}{\cos 40^\circ + 0.827 \dots \times \sin 40^\circ}$ $T = 73.55939193 = 74$ (2

(allow sign errors) OR Eliminates R to solve for T			sf
Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
Solves for T	AO1.1b	A1	
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	
Total 12 marks			

Q9.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.

B1: Diagram with one error or omission.

B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20^\circ = 30 \times 9.8$

$(R =) 30 \times 9.8 - 150 \sin 20^\circ$

$= 242.69 \dots$

M1: Resolving vertically to obtain a three term equation, with R , $150 \sin$ or $\cos(20^\circ$ or $70^\circ)$ and $30g$ oe.

A1: Correct equation. Allow g instead of 9.8 .

M1A1

$= 243 \text{ N (to 3sf)}$

A1: AG Correct final answer having seen either 2^{nd} or 3^{rd} or both line of solution.

A1

3

(iii) $(F =) 0.4 \times 242.7 = 97.1 \text{ N}$

M1: Use of $F = \mu R$ or $F \leq \mu R$.

A1: Correct final answer without an inequality. Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20^\circ - 97.08$

M1: Three term equation of motion with $30a$, $150 \sin$ or $\cos(20$ or $70^\circ)$ and their friction from (a)(iii). Condone incorrect signs.

A1: Correct equation.

M1A1

$$a = \frac{150 \cos 20^\circ - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a .

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWR 1.46.

A1

4

(b) $R = 30 \times 9.8 - T \sin 20^\circ$

B1: Correct normal reaction in terms of T .

B1

$$F = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

B1: Correct friction in terms of T

B1

$$T \cos 20^\circ = 0.4 (30 \times 9.8 - T \sin 20^\circ)$$

*M1: Resolving tension horizontally and equating to F ,
provided that F is in terms of T .*

A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20^\circ + 0.4 \sin 20^\circ} = 109 \text{ N}$$

A1: Correct tension. AWRT 109.

A1

5

(c) The same

B1: The same.

Use of $g = 9.81$ gives acceptable final answers.

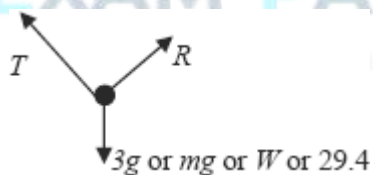
B1

1

[17]

Q10.

(a)



*B1: Correct force diagram with arrows
and sensible labels.*

If R is shown as vertical award B0.

If F is included, award B0

*Accept a reflection of the diagram in a
vertical line.*

*Ignore components if shown with a
different notation, eg dotted lines.*

B1

1

(b) ($R =$) $3g \cos 60$

M1: Resolving perpendicular to the slope. Must see $\cos 60^\circ$ or $\sin 30^\circ$ or $\cos 30^\circ$ or $\sin 60^\circ$ and $3g$ or 29.4 .

NOTE: $\frac{3g}{2} = 14.7$ or equivalent without the use of a trig term scores M0.

M1

(R =) 14.7 **AG**

A1: Correct value from correct working.
 NOTE: If candidates use $g = 9.81$, deduct one mark here. If candidates obtain 14.7 from 14.715 they will have used $g = 9.81$.
 Note: "R =" does not need to be seen.

A1

2

(c) (T =) $3g \sin 60^\circ$

M1: Resolving parallel to the slope. Must see $\cos 60^\circ$ or $\sin 30^\circ$ or $\cos 30^\circ$ or $\sin 60^\circ$ and $3g$ or 29.4 .

M1

(T =) 25.5

A1: Correct value. AWR 25.5 or truncation to 25.4.
 NOTE: If candidates use $g = 9.81$ again, do not penalise. Use of $g = 9.81$ gives 25.5 for the tension.
 Note: "T =" does not need to be seen.

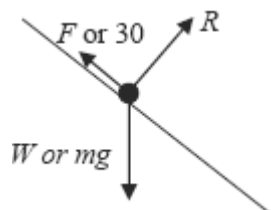
A1

2

[5]

Q11.

(a)



B1: Correct force diagram with arrows and labels.

B1

1

(b) $30 = mg \sin 42^\circ$

M1: Two term equation by resolving parallel to the slope. Must include g.

A1: Correct equation.

M1A1

$$m = \frac{30}{9.8 \sin 42^\circ} = 4.57$$

A1: Correct value for m.

A1

3

(c) $(R =) mg \cos 42^\circ = 33.3 \text{ N}$

M1: Resolving perpendicular to the slope to find R. Must include g.

A1: Correct value.

M1A1F

2

(d) $30 = \mu R$

$$\mu = \frac{30}{R} = 0.900$$

M1: Use of $F = \mu R$

M1

M1: Solving for μ .

M1

A1F: Correct value for μ . Accept 0.9.

A1F

3

Use of $g = 9.81$

(b) 4.57 B1

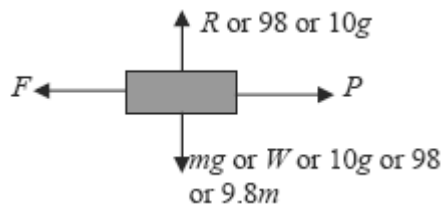
(c) 33.3 B1

(b) 0.900 B1

[9]

Q12.

(a)



B1: Correct force diagram with arrows and labels.

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 10 kg for the weight.

Note: Do not accept μR or $0.5R$ for F .

B1

1

(b) (i) $(R = 10 \times 9.8 =) 98 \text{ N}$

B1: Correct normal reaction. Accept 10g.

No need to see the letter R or working.

B1

1

(ii) $(F \leq) 0.5 \times 98$

$(F \leq) 49$

B1: Correct maximum value for friction.

Accept 5g.

No need to see the letter F or any working. Ignore any inequalities.

For FT, must be 0.5 of candidate's answer to (b)(i).

B1F

1

(iii) $(F =) 30 \text{ N}$

B1: Correct friction. Allow – 30.

B1

1

(c) $80 - 49 = 10a$

M1: Three term equation motion, containing 80, candidate's 49 and $10a$ (not $10ga$) in any combination.

A1F: Correct equation including signs.

M1A1F

$a = 3.1 \text{ m s}^{-2}$

A1F: Correct acceleration.

FT candidate's answer to (b)(ii).

A1F

3

Allow use of $g = 9.81$

(b) (i) 98.1

B1

(b) (ii) 49.05 or 49.1 or 49

B1

(c) 3.095 or 3.09 or 3.1

M1A1A1

[7]

Q13.

- (a) Resultant Force = 300×2.2
= 660 N AG

B1: Correct value from correct multiplication.

B1

1

- (b) $P - 400 = 660$
 $P = 1060$

M1: Three term equation of motion

M1

A1: Correct value for P

A1

2

- (c) $23 = 12 + 2.2t$

M1: Use of a constant acceleration equation to find t .

M1

A1: Correct equation

A1

$$t = \frac{23 - 12}{2.2} = 5 \text{ s}$$

A1: Correct time

A1

3

[6]

Q14.

- (a) (i) $20 \times 9.8 = R + 60 \sin 30^\circ$

M1: Equation or expression for normal reaction with mg or $20g$ or 196 and $60 \sin 30^\circ$ or $60 \cos 30^\circ$.

A1: Correct equation or expression with correct signs.

M1A1

$$(R =) 20 \times 9.8 - 60 \sin 30^\circ = 166 \text{ N} \quad \text{AG}$$

*A1: Correct value from correct working.
Must be positive.*

Don't penalise use of $g = 9.81$ if already done earlier on script. Should still get 166, but from 166.2.

A1

3

(ii) $166\mu = 60 \cos 30^\circ$

*M1: Use of $F = \mu R$, with $R = 166$ or 166.2. Do not allow inequalities here.
M1: Resolving horizontally with $\cos 30^\circ$ or $\sin 30^\circ$ oe*

A1: Correct equation

Examples:

$$166\mu = 60 \quad \text{M1M0A0}$$

$$166\mu = -60 \cos 30^\circ \quad \text{M1M1A0}$$

M1M1A1

$$\mu = \frac{60 \cos 30^\circ}{166}$$

$$= 0.313$$

A1: Correct coefficient of friction.

A1

4

(b) $20 \times 0.8 = T \cos 30^\circ - 0.313(20 \times 9.8 - T \sin 30^\circ)$

B1: $20g - T \sin 30^\circ$ oe seen.

M1: Three term equation of motion, where normal reaction is dependent on T .

A1F: Correct equation

B1M1A1F

$$T = \frac{20 \times 0.8 + 0.313 \times 20 \times 9.8}{\cos 30^\circ + 0.313 \sin 30^\circ} = 75.6 \text{ N}$$

dM1: Solving for T including factorisation.

A1F: Correct tension.

AWRT 75.6

Follow through incorrect values of μ from part (a).

Don't penalise use of $g = 9.81$ if already

done earlier on script. Should get 75.7.
Allow 75.8 if intermediate values rounded.

5

[12]

Q15.

(a)

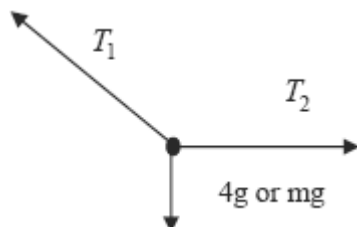


Diagram with three forces, labels and arrow heads. Different variables must be used for each tension

B1

1

(b) $T_1 \sin 30^\circ = 4 \times 9.8$

Two term equation from resolving vertically.
Must see a sin or cos term for M1

M1

$$T_1 = \frac{4 \times 9.8}{\sin 30^\circ} = 78.4 \text{ N}$$

AG

Correct equation

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A1

Correct tension from correct working. All rights reserved.

A1

3

(c) $T_2 = 78.4 \cos 30^\circ = 67.9 \text{ N}$

Two term equation from resolving horizontally.

M1

Correct tension.

A1

2

[6]

Q16.

(a) $2 \times 9.8 - T = 2 \times 0.9$

Three term equation of motion for B

M1

Correct equation

A1

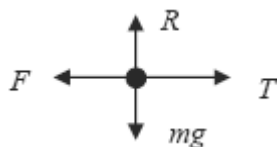
$$T = 19.6 - 1.8 = 17.8 \text{ N}$$

Correct tension

A1

3

(b)



Correct force diagram with labels and arrows

B1

1

(c) $R = 3 \times 9.8 = 29.4 \text{ N}$

Calculation of normal reaction force.

B1

1

(d) $17.8 - F = 3 \times 0.9$

Three term equation of motion for A.

M1

$$F = 17.8 - 2.7 = 15.1 \text{ N}$$

Correct friction

A1

2

(e) $15.1 = 29.4\mu$

Use of $F = \mu R$

M1

$$\mu = \frac{15.1}{29.4} = 0.514$$

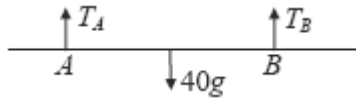
Correct value for μ

A1

2

Q17.

(a)



B1

1

(b) Taking moments about A

$$2.1 \times 40g = T_B \times 4$$

B1 for 2.1

M1 B1

$$T_B = 21g$$

A1

3

(c) Resolve vertically $T_A + T_B = 40g$

M1

$$T_A = 19g \text{ or } 186 \text{ N}$$

A1

2

(d) Gravitational force acts through mid point of the rod

E1

1

[7]

Q18.

(a) $320 = \frac{1}{2} \times a \times 80^2$

*Use of constant acceleration equation
with $u = 0$*

M1

Correct equation

A1

$$a = \frac{2 \times 320}{80^2} = 0.1 \text{ m s}^{-2}$$

AG Correct acceleration from correct working

A1
3

(b) $v = 0 + 0.1 \times 80$

*Use of constant acceleration equation
with $u = 0$*

M1

$= 8 \text{ m s}^{-1}$

Correct velocity

A1
2

(c) $L - 450 \times 9.8 = 450 \times 0.1$

Three term equation of motion

M1

Correct equation

A1

$L = 45 + 4410$

$= 4455$

$= 4500 \text{ N (to 3sf)}$

AG Correct force

A1
3

(d) 4410 N

Correct force

B1
1

[9]

Q19.

(a) (i) $T = 0.6 \times 9.8 = 5.88 \text{ N}$ or $0.6g$

B1
1

(ii) $\text{Force} = 2T = \downarrow 11.76 \text{ N}$ or 11.8 N or $1.2g$

B1
2

Magnitude

B1

direction

B1

(b) (i) $Q: 0.8g - T = 0.8a$

M1

$T - 0.6g = 0.6a$
 either equation

A1

$0.2g = 1.4a$

A1

alt M1A1 if solving for T , full method for solving,
 A1 accurate attempt

m1

$a = 1.4a$

A1

$T = 6.72\text{N}$
 CAO
S.C whole string.
 To find a : $0.2g = 1.4a$ (M1)
 $a = 1.4$ A1
 To find T : (M1A1)
 CAO

A1

6

(ii) Force = $2T = 13.44\text{N}$

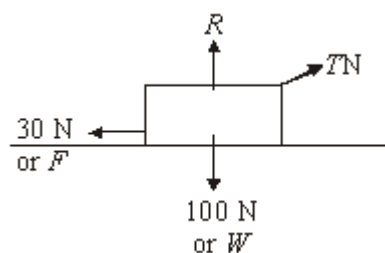
B1

1

[10]

Q20.

(a)



B1
1

(b) $F = \mu R, \quad 30 = 0.5 R$

M1

$$R = 60\text{N}$$

A1
2

(c) Constant speed $\Rightarrow$ no acceleration / forces

M1

$$\text{balance} \therefore 30\text{N} = T_x$$

A1
2

(d) $T_y + R = 100$

M1

$$T_y = 40$$

A1

$$T = \sqrt{(30^2 + 40^2)}$$

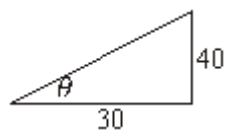
Use of a vertical component of T

M1

$$T = 50$$

A1
4

(e)



$$\tan \theta = \frac{40}{30}$$

M1

A1ft

$$\theta = 53.1^\circ \text{ Accept } 53^\circ$$

CAO

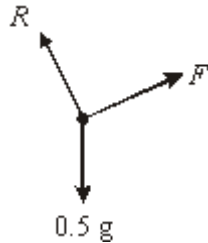
A1

3

[12]

Q21.

(a)



Accept W or mg (or 4.9) for weight
 Arrows and labels needed
 Can replace W with 2 correct components

B1

1

(b)

$$F = 0.5g \sin 16^\circ$$

component of weight attempted

All correct including signs

M1

A1

$$F = 1.35 \text{ N}$$

CAO

A1

3

(c)

$$R = 0.5g \cos 16^\circ$$

component of weight attempted

M1

$$R = 4.71 \text{ N}$$

CAO

(b) and (c): For resolving horizontal/vertical M1
 for each equation, A1 one equation correct

F A1 R A1

A1

2

(d) $F \leq \mu R$
 $1.35 \leq \mu \times 4.71$

Use of candidate's R, accept equals sign

M1

$\mu \geq 0.287$

CAO inequality required

A1

2

[8]

Q22.

(a) $P = 5 + 8\cos 60^\circ$

Both relevant forces, component of 8N attempted

M1

All correct

A1

$P = 9$

CAO

A1

3

(b) $Q = 8\cos 30^\circ$

Component of 8N attempted

M1

$Q = 6.93$ or $4\sqrt{3}$

AWRT 6.93

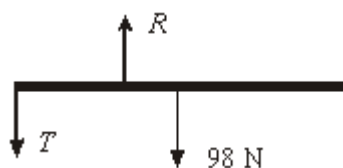
A1

2

[5]

Q23.

(a)



correct force diagram, with labels and arrows.

B1

1

(b) $2T = 0.5 \times 98$

moment equation

M1

$T = 24.5 \text{ N}$

correct equation

A1

AG

*correct positive value for the tension from
correct working*

A1

3

(c) (i) $2 \times 2 \times 24.5 = 3 \times 9.8 \times m + 0.5 \times 98$

tension doubled

B1

moment equation

M1

$$m = \frac{98 - 49}{3 \times 9.8} = \frac{5}{3} = 1.67 \text{ kg (to 3 sf)}$$

correct equation

A1

correct mass

A1

4

OR

$2 \times 24.5 = 3 \times 9.8m$

for equation

(M1A1)

$$m = \frac{49}{29.4} = \frac{5}{3} = 1.67 \text{ kg}$$

for finding m

(M1A1)

(ii) $R = 24.5 \times 2 + 98 + \frac{5}{3} \times 9.8 = 163 \text{ N}$

considering vertical equilibrium with 3 terms

M1

correct equation

A1

correct reaction

A1

3

must be consistent with 3(c)(i) if awarding accuracy marks

- (d) This allows the centre of mass to be placed at the centre of the rod for the moment calculations.

correct explanation

B1

1

[12]

Q24.

(a) $200 \sin 30^\circ = T \sin 45^\circ$

$$T = \frac{200 \sin 30^\circ}{\sin 45^\circ} = 141 \text{ N}$$

Resolving horizontally

M1

Correct equation

A1

AG; correct T from correct working

A1

3

(b) $200 \cos 30^\circ + 141 \cos 45^\circ + R = 500 \times 9.8$

Resolving vertically with four terms

M1

Correct values

A1

Correct signs

A1

$R = 4630 \text{ N}$

Correct R

A1

4

[7]

Q25.

- (a) No air resistance / Only gravity or weight

B1: Acceptable assumption

B1

1

- (b) (i) $0.2 \times 8 = 0.2 \times 9.8 - R$

M1: Three term equation of motion

A1: Correct equation

M1A1

$$R = 0.36 \text{ N}$$

A1: Correct magnitude of the resistance force

A1

3

- (b) (ii) **Increases** as the speed increases

B1: Correct explanation

B1

1

- (c) (i) $\pm 9.8 \text{ ms}^{-2}$

B1: CAO

B1

1

- (ii) **Decreases** towards zero

B1: Correct explanation

B1

1

[7]

Q26.

(a) $R = 2 \times 9.8, \quad F = \frac{1}{7} \times 2 \times 9.8 = 2.8 \text{ N}$

*CAO M1: full method with 2×9.8 ,
accept inequality*

M1A1

2

(b) (i) $P = 1, \quad F = 1\text{N}$

B1

1

(ii) $P = 5, \quad F = 2.8$

Used

B1

$$5 - 2.8 = 2a$$

M1: 3 terms, with forces subtracted

M1A1

$$a = 1.1 \text{ ms}^{-2}$$

ft one slip

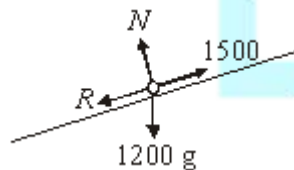
A1F

4

[7]

Q27.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$

M1: 3 terms with component attempted

($\alpha = 4.096$) – 1 each error

M1A2

$$R = 660$$

AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $-\frac{1500}{1200}$

Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1

2

(ii) $0 = 15 + (-1.25) t$

velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec}$ (27.27)
ft one slip

A1F

2

[11]

Q28.

(a) $5 = 10 \cos \theta^\circ$

Component attempted

M1

$\theta = 60$

AG

A1

2

(b) $Q = 10 \sin \theta$

Component attempted

M1

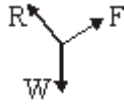
$Q = 5\sqrt{3} \text{ N} = 8.66 \text{ N}$

Accept either form

A1

Q29.

(a)



Three forces labelled & with arrows, W , mg or 35 g vertical (or 2 components of W), R and F perpendicular (ignore pairs of components of existing forces)

B1

1

(b) $R = 35g \times \cos 25^\circ$

Component attempted & g present for M1

M1A1

$R = 311\text{ N}$

($R = 310.86$) accept AWRT 311

A1F

3

(c) $F = 35g \times \sin 25^\circ$

Component attempted & g present for M1 & acceleration zero

M1A1

($= 144.96$)

$F = \mu R, 144.96 = \mu \times 310.86$

Use of friction law with candidate's values, must have tried to find F

M1

$\mu = 0.466$

ft R and F , provided $\mu > 0$

M1A0 if $F < \mu R$ used

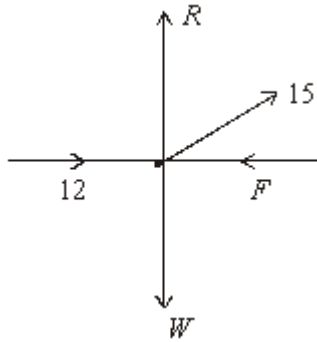
SC accept use of $\mu = \tan \theta$

A1F

4

Q30.

(a)



any 3 correct, including arrows and labels

B1

all correct, including arrows and labels

B1

2

accept mg for W

accept either 15 N **or** components, but **not** both

(b) $F = 12 + 15 \cos 30^\circ$

for M1, both terms on RHS and sin/cos

M1A1

$= 12 + 12.99$

$F = 25 \text{ N}$

CAO

A1

3

(c) $25 = \frac{1}{3} R$

M1A0 if inequality used

M1

$R = 75 \text{ N}$

AWRT; ft F ; accept use of c 's F from (b);

accept 0.333 for μ

A1F

2

(b) $R + 15 \cos 60^\circ = W$

3 terms and attempt at component

$W_g \Rightarrow$ M1A0A0

M1

$$75 + 7 \frac{1}{2} = W$$

ft R

A1F

$$W = 82 \frac{1}{2} \text{ N}$$

AWRT 82.5, ft R or one slip

A1F

3

[10]

Q31.

- (a) (i) **trailer:** $\pm S + 100 = 250 \times 0.5$
*M1 for 3 terms, A1 all correct, allow $\pm S$
 g included with mass $\Rightarrow$ M1A0*

M1A1

$$S = 25\text{N}$$

*ft one error (including g present) provided $S > 0$
 If S negative, accept sign changed for answer*

A1F

3

- (ii) **car + trailer:**

$$500 + 100 + R = 150 \times 0.5$$

*All terms for M1, including appropriate mass
 g included with mass $\Rightarrow$ M1 A0 A1*

or: car:

$$R + 500 - S = 1250 \times 0.5$$

-1 each error; ft S if used

A2, 1, 0

$$R = 150 \text{ N}$$

*ft one slip, for $R > 0$
 If R consistently in wrong direction, need
 explanatory comment re direction for A1,
 otherwise M1 A1 A0 A0 ($R < 0$)*

A1F

4

(b) $T = 100 \text{ N}$
 CAO

B1

1

[8]

Q32.

(a) $150g - T = 150 a$
Three term equation of motion for tank.

M1

$T - 200g \sin 30^\circ = 200 a$
Correct equation

A1

$150g - 200a - 200g \sin 30^\circ = 150 a$
Three term equation of motion for trolley

M1

Correct equation

A1

$a = \frac{150g - 200g \sin 30^\circ}{350} = 1.4 \text{ ms}^{-2}$

Solving for a

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Correct equation for a

A1

ag *Correct a from correct working.*

A1

7

(b) $T = mg$
Equations of motion for both bodies.

M1

$T = 200 g \sin 30^\circ$
Correct equation for tank.

Correct equation for trolley

A1

A1

$$m = 200 \sin 30^\circ = 100 \text{ kg}$$

Correct mass

A1

4

[11]

Q33.

(a) $T - 800 \times 9.8 = 800 \times 0.2$

Three term equation of motion

M1

Correct equation

A1

$$T = 7840 + 160 = 8000 \text{ N}$$

ag Correct tension

A1

3

(b) $T = 800 \times 9.8 = 800 \times (0.2)$

Three term equation of motion

M1

$$T = 7840 - 160 = 7680 \text{ N}$$

Correct tension

A1

2

(c) $T = 800 \times 9.8 = 7840 \text{ N}$

Correct tension

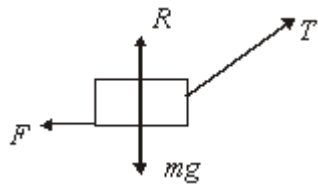
B1

1

[6]

Q34.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 30^\circ = 200 \times 9.8$

Three term equation from resolving vertically

M1

$$R + 0.5 T = 1960$$

$$R = 1960 - 0.5T$$

Correct equation

A1

ag Correct result from correct working

A1

3

(c) $F = T \cos 30^\circ$

Resolving horizontally

M1

$$T \cos 30^\circ = 0.6(1960 - 0.5T)$$

Use of $F = \mu R$

M1

$$T (\cos 30^\circ + 0.3) = 1176$$

Correct equation

A1

Solving for T

m1

$$T = \frac{1176}{(\cos 30^\circ + 0.3)} = 1010 \text{ N}$$

Correct T

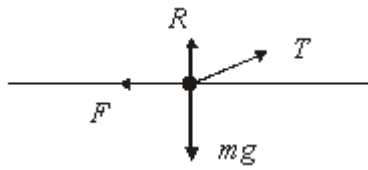
A1

5

[9]

Q35.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

$$R = 490 - T \sin 40^\circ$$

Three term equation of motion

M1

Correct equation

A1

Correct expression for R

Use of $F R = \mu$

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$

Use of $F = \mu R$

M1

$$= 294 - 0.6 T \sin 40^\circ$$

ag *Correct result from correct working*

A1

2

(d) $T \cos 40^\circ - (294 - 0.6 T \sin 40^\circ) = 50 \times 0.5$

$$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

Correct T

A1

4

[10]



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